Gaining Root Access, Creating Linux User and Group Management

What is Root Access?

- Root is the superuser account in Linux
- Has unrestricted access to all commands and files
- Identified by User ID (UID) 0
- Essential for system administration tasks

Methods to Gain Root Access

1. Direct root login (typically disabled for security)
2. Using the **`su`** command
3. Using **`sudo`** for specific commands
4. Using **`sudo su`** or **`sudo -i`** for a root shell

'su' Command

- In Linux, the **su** (***short for substitute user***) command allows you to switch to another user account, including the superuser (root), during your current session.
- By default, if no username is specified, su assumes you want to switch to the root user.
- This command requires the password of the user you are switching to.

Basic Usage

Switch to another user (without root privileges):

```
su username
```

After entering the password of the specified user, you will have the same privileges as that user.

Switch to the root user:

```
su -
```

You will be prompted to enter the root user's password. Once authenticated, you'll gain root privileges and switch to the root user's environment (home directory, environment variables, etc.).

With or Without a Dash (-)

su username: Switches to the specified user but retains the current user’s environment (e.g., the working directory remains the same).

su - username: Switches to the specified user and loads that user’s full environment (home directory, environment variables, etc.).

su - (without specifying a username): Switches to the root user and loads the root user's full environment.

Important Point

- **~ (Tilde):** Represents the current user's home directory.
- **/ (Forward Slash):** Represents the root directory of the Linux file system.

‘sudo’ Command

- The sudo (Superuser DO) command in Linux is used to execute commands with superuser (root) privileges.
- It allows a permitted user to execute a command as the superuser or another user, as specified by the security policy.
- Here are some basic tasks that can be done using the **sudo** command. These tasks are essential for managing a Linux system with elevated privileges:

Changing User Password

```
sudo passwd username
```

Use this to change the password of any user, including your own.

Installing Packages

```
sudo apt-get install package-name (for Debian/Ubuntu-based distros)
sudo yum install package-name (for RHEL/CentOS-based distros)
```

This command allows you to install software packages.

Shutting Down or Rebooting the System

```
sudo shutdown now
sudo reboot
```

Used for shutting down or rebooting the system immediately.

Important Notes

1) **Sudo prompts for your password:** When you run a **sudo** command, it asks for your user password (not the root password). You need to provide it to continue.

2) **Be cautious:** Since **sudo** grants administrative privileges, be careful when using it. Running commands with **sudo** can affect important system settings or files if done incorrectly.
3) **Timeout:** Once you've entered your password, **sudo** won't ask for it again for a certain amount of time (usually 15 minutes), making it easier to run several administrative commands in succession.

Practice Questions

1) Try switching to root using `su -`. What happens?

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo su
[sudo] password for abdullah-sohail:
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# whoami
root
```

2) Use `sudo` to view the contents of `/etc/shadow`

```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# sudo cat /etc/shadow
root:*:20305:0:99999:7:::
daemon:*:20305:0:99999:7:::
bin:*:20305:0:99999:7:::
sys:*:20305:0:99999:7:::
sync:*:20305:0:99999:7:::
games:*:20305:0:99999:7:::
man:*:20305:0:99999:7:::
lp:*:20305:0:99999:7:::
mail:*:20305:0:99999:7:::
news:*:20305:0:99999:7:::
uucp:*:20305:0:99999:7:::
proxy:*:20305:0:99999:7:::
www-data:*:20305:0:99999:7:::
backup:*:20305:0:99999:7:::
list:*:20305:0:99999:7:::
irc:*:20305:0:99999:7:::
_apt:*:20305:0:99999:7:::
nobody:*:20305:0:99999:7:::
systemd-network:!*:20305::::::
systemd-timesync:!*:20305::::::
dhcpcd:!:20305::::::
messagebus:!:20305::::::
syslog:!:20305::::::
systemd-resolve:!*:20305::::::
uuidd:!:20305::::::
usbmux:!:20305::::::
tss:!:20305::::::
systemd-oom:!*:20305::::::
```

3) Create a file in `/root` directory using `sudo`

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo touch /root/myfile.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo ls /root
myfile.txt  snap  testfile.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

4) Switch the user from the current user's home directory to root but user current directory didn't change.

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ pwd
/home/abdullah-sohail
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ su
Password:
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# whoami
root
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# pwd
/home/abdullah-sohail
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail#
```

5) Change the password for your user account.

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ passwd
Changing password for abdullah-sohail.
Current password:
New password:
BAD PASSWORD: The password fails the dictionary check - it does not contain enough DIFFERENT characters
New password:
BAD PASSWORD: The password fails the dictionary check - it does not contain enough DIFFERENT characters
New password:
Retype new password:
passwd: password updated successfully
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

6) Change the password for another user (requires root or sudo privileges).

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo passwd ahs
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
```

7) Update the package lists (Debian/Ubuntu).

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://pk.archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://pk.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://pk.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,494 kB]
Get:6 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,800 kB]
Get:7 http://pk.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [331 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [240 kB]
Get:9 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [174 kB]
Get:10 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.5 kB]
Get:11 http://pk.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,735 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [9,932 B]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2,568 kB]
Get:15 http://pk.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [630 kB]
Get:16 http://pk.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:17 http://pk.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,558 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [596 kB]
Get:19 http://pk.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [316 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:21 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [961 kB]
Get:22 http://pk.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:23 http://pk.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [32.7 kB]
Get:24 http://pk.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7,044 B]
Get:25 http://pk.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
```

8) Install a specific package (e.g., **vim**) on Debian/Ubuntu.

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo apt install vim
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libsodium23 vim-runtime
Suggested packages:
  ctags vim-doc vim-scripts
The following NEW packages will be installed:
  libsodium23 vim vim-runtime
0 upgraded, 3 newly installed, 0 to remove and 205 not upgraded.
Need to get 9,324 kB of archives.
After this operation, 42.0 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 libsodium23 amd64 1.0.18-1ubuntu0.24.04.1 [161 kB]
Get:2 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 vim-runtime all 2:9.1.0016-1ubuntu7.9 [7,281 kB]
Get:3 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 vim amd64 2:9.1.0016-1ubuntu7.9 [1,881 kB]
Fetched 9,324 kB in 6s (1,576 kB/s)
Selecting previously unselected package libsodium23:amd64.
(Reading database ... 154505 files and directories currently installed.)
Preparing to unpack .../libsodium23_1.0.18-1ubuntu0.24.04.1_amd64.deb ...
Unpacking libsodium23:amd64 (1.0.18-1ubuntu0.24.04.1) ...
Selecting previously unselected package vim-runtime.
Preparing to unpack .../vim-runtime_2%3a9.1.0016-1ubuntu7.9_all.deb ...
Adding 'diversion of /usr/share/vim/vim91/doc/help.txt to /usr/share/vim/vim91/doc/help.txt.vim-tiny by vim-runtime'
Adding 'diversion of /usr/share/vim/vim91/doc/tags to /usr/share/vim/vim91/doc/tags.vim-tiny by vim-runtime'
Unpacking vim-runtime (2:9.1.0016-1ubuntu7.9) ...
```

9) Shut down the system after 10 minutes.

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo shutdown +10
Shutdown scheduled for Sun 2026-03-01 12:33:55 PKT, use 'shutdown -c' to cancel.
```

User Management

User and group management is a fundamental aspect of Linux system administration. Following portion will guide you through the concepts and practical applications of managing users and groups in a Linux environment.

Key Concepts:

1. **Users:** Individual accounts on a Linux system.
2. **Groups:** Collections of users with shared permissions.
3. **Permissions:** Read, write, and execute access controls for files and directories.
4. **Superuser (root):** The administrative user with unrestricted access.

Part 1: Basic User Management

- User information is stored in ***/etc/passwd***
- User passwords are stored (hashed) in ***/etc/shadow***
- Each user has a unique User ID (UID)

Commands:

1. **useradd:** Add a new user

2. **passwd:** Set or change user password
3. **usermod:** Modify user account
4. **userdel:** Delete a user account
5. **id:** Display user and group IDs

Examples:

1. Create a new user:

```
sudo useradd -m ali
```

-m (create home directory):

The `-m` option tells `useradd` to *create a home directory* for the new user. By default, Linux will place the home directory under `/home` (e.g., `/home/ali` for the user `ali`). If this option is omitted, no home directory will be created, and the user will not have a dedicated space to store files.

2. Set a password for the new user:

```
sudo passwd ali
```

3. Add a user to an existing group:

```
sudo usermod -aG sudo ali
```

usermod:

The ***usermod*** command is used to modify user account information on Linux/Unix systems. This command allows administrators to change properties like group memberships, login shell, home directory, etc.

-a (append):

The ***-a*** option stands for append. It tells the usermod command to add the user (ali) to the specified group (sudo) without removing the user from other groups. Without -a, the user would be removed from all other groups and added only to the specified group.

-G (group):

The -G option is used to specify the group(s) to which the user should be added. Multiple groups can be specified, separated by commas (e.g., -G group1, group2, group3).

sudo (group):

This is the group name. In this command, sudo is a special administrative group that provides members the ability to use the sudo command and gain superuser privileges temporarily.

4. Delete a user:

```
sudo userdel -r ali
```

-r:

The `-r` option stands for **remove**.

Practice Questions:

1. Create a new user named "alice" with a home directory.
2. Set a password for alice.
3. Create another user named "bob" without a home directory.
4. Use the id command to display information about alice and bob.

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo useradd -m alice
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo passwd alice
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
passwd: password updated successfully
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo useradd ahs
useradd: user 'ahs' already exists
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ id alice
uid=1002(alice) gid=1002(alice) groups=1002(alice)
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ id ahs
uid=1001(ahs) gid=1001(ahs) groups=1001(ahs)
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

Part 2: Basic Group Management

- Group information is stored in /etc/group
- Each group has a unique Group ID (GID)
- Users can belong to multiple groups

Commands:

1. **groupadd:** Create a new group
2. **groupdel:** Delete a group
3. **groupmod:** Modify group properties
4. **groups:** Display group membership
5. **newgrp:** Change current group ID

Examples:

1. Create a new group:

```
sudo groupadd developers
```

2. Add a user to a group:

```
sudo usermod -aG developers ali
```

3. Remove a user from a group:

```
sudo gpasswd -d ali developers  OR sudo deluser ali developers
```

4. Delete a group:

```
sudo groupdel developers
```

Basic Commands to View Groups

View all groups

```
getent group

sudo:x:27:user1,user2
```

View groups for current user

```
groups
```

View groups for specific user

```
groups username
```

Understanding Group File (/etc/group)

Each line in `/etc/group` contains four fields separated by colons:

```
group_name:password:GID:user_list
```

View the file:

```
cat /etc/group
```

Field Explanation:

1. **group_name**: The name of the group
2. **password**: Usually x (actual password stored in /etc/gshadow)
3. **GID**: Group ID number
4. **user_list**: List of users who are members of the group

Advanced Group Information

View primary group for a user

```
id -gn username
```

View all groups with GIDs

```
id username
```

Check if a user is in a specific group

```
groups username | grep groupname
```

Practice:

1. Create a new group called "project_a".
2. Add ali and bob to the "project_a" group.
3. Create another group called "project_b" and add only bob to it.
4. Use the groups command to verify group memberships for ali and bob.

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo groupadd project_a
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo usermod -aG project_a ahs
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo usermod -aG project_a alice
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo groupadd project_b
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo usermod -aG project_b ahs
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ groups ahs
ahs : ahs project_a project_b
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ groups alice
alice : alice project_a
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```